
GPS precision & accuracy — CYC Wed Spring Race 6 (2026-07-01)

GPS-quality review for every boat with a track in Race 6 — the **three SailFrames units** plus **six owner-supplied tracks** (Vakaros, phone apps, GPS loggers). Race 003bef01 · sailframes.com/race.html?race=003bef01 .

Boat	Source	Device / module	Rate	Data
Katü	SailFrames B2	Quectel LC29HEA	10 Hz	✓ 190,889 fixes
Wizard	SailFrames E3	Waveshare LG290P	10 Hz	✓ 88,764 fixes
Doc Buck	SailFrames E4	Waveshare LG290P	10 Hz	✓ 103,898 fixes
Amigo	SailFrames B1	Quectel LC29HEA	—	✗ no data — did not record
RockIt 2.0	Vakaros	Atlas race processor	5 Hz	✓ 42,891 pts
Uproarious	GPX logger	(reports sats)	1 Hz	✓ 12,418 pts
MASHNEE	GPX logger	—	1 Hz	✓ 10,063 pts
Eagle (McLean)	GPX logger	—	0.1 Hz	✓ 582 pts (sparse)
Agora	phone	Sensor Logger app	1 Hz	✓ 4,454 pts
Never Settle	phone	Sensor Logger app	1 Hz	✓ 9,812 pts

⚠ **These are heterogeneous devices, antennas, mounts and log rates** — not a controlled A/B. The SailFrames units ran **RTK disabled** (plain GPS/SBAS, fix 1–2, no cm-level RTK). Treat cross-device numbers as indicative, not a ranking of silicon.

TL;DR

- **Logging quality is excellent on all three that recorded:** solid 10 Hz, no gaps > 1 s, ≥ 30 sats.
- **The LG290P (E3/E4) has the stronger front end** — ~34 sats vs ~30 and much lower, steadier HDOP (~0.30 vs ~0.45) than the LC29HEA (B2).
- **Position precision** (scatter while sitting still): **E3 ≈ 0.9 m CEP50 is the cleanest result**; B2 and E4 land ~2 m, but E4's number is inflated by mooring swing, not receiver noise.
- **“Accuracy” (closeness to truth) is not measured here** — there is no survey ground-truth, and the receivers' own `hacc` is self-reported (and, on the LG290P, essentially a fixed 1.30 m floor). Don't rank the receivers on `hacc` .
- **B1/Amigo captured nothing** (a single fix at 42.8 m / 4 sats). It didn't record the race — a fleet-readiness failure, not a GPS-quality data point.
- **Across all sources, log rate spans 100×** (10 Hz → 0.1 Hz). Only the SailFrames units (10 Hz) and RockIt's Vakaros Atlas (5 Hz) are fine-grained enough for per-tack / start-line tactical work; the 1 Hz phone/GPX logs are coarse and Eagle's 0.1 Hz is a bare track.

- **Vakaros Atlas (RockIt)** shows the **tightest scatter (~0.1 m)** — but that’s a filtered/smoothed race-processor output, not raw GNSS, so it isn’t comparable to the raw receivers.
- **Phone GPS (Sensor Logger)** is the **least accurate** — 2–8 m self-reported and spiky (p95 to 14 m); fine for a rough track, not for close-quarters.

All 9 sources at a glance

Precision = position scatter over each boat’s **stillest window** (longest run < 0.5 kt). “n/a” = no clean still window in that log (device never sat still, or the log has no speed field to detect it). Self- hacc only where the device reports it.

Boat	Source	Rate	Stillest window	CEP50	DRMS	Self- hacc (med / p95)
Wizard (E3)	SailFrames LG290P	10 Hz	585 s	0.88 m	1.29 m	1.3 m (floor)
Katü (B2)	SailFrames LC29HEA	10 Hz	6428 s	2.09 m	2.36 m	2.2 / 3.6 m
Doc Buck (E4)	SailFrames LG290P	10 Hz	318 s	2.07 m†	2.36 m†	1.3 m (floor)
RockIt 2.0	Vakaros Atlas	5 Hz	49 s	0.06 m*	0.10 m*	—
Uproarious	GPX logger	1 Hz	985 s	0.82 m	3.91 m	—
Never Settle	phone (Sensor Logger)	1 Hz	4005 s	1.05 m	5.62 m	2.1 / 14.2 m
Agora	phone (Sensor Logger)	1 Hz	n/a	—	—	4.8 / 8.1 m
MASHNEE	GPX logger	1 Hz	n/a	—	—	—
Eagle (McLean)	GPX logger	0.1 Hz	n/a	—	—	—

* **Vakaros Atlas ~0.1 m is filtered/smoothed** race-processor output, not raw GNSS noise — do not compare it directly to the raw receivers. † E4’s number is inflated by mooring swing (see below), not receiver noise.

What this says about picking a tracker: - For **tactical analysis** (starts, per-tack VMG, close-quarters), you need ≥ 5 Hz: only the **SailFrames (10 Hz)** and **Vakaros (5 Hz)** qualify. The 1 Hz phone/GPX logs and Eagle’s 0.1 Hz miss the fast dynamics. - Among the **raw** receivers, the **LG290P (E3)** gives the best precision/rate balance (~0.9 m at 10 Hz); the LC29HEA (B2) is ~2 m. - **Phone GPS is the weakest** — usable for a coarse track and replay, spiky at the metres level, not for boat-length decisions.

What “precision” and “accuracy” mean here

- **Precision (repeatability)** — how much the reported position wanders while the antenna is physically fixed. We measure it from each boat’s **stillest window** (longest run under 0.5 kt) as position standard deviation, DRMS, and CEP50/CEP95. This is a real, measurable number — but

it's an **upper bound**: it also captures real antenna motion (mooring swing, float movement), so the true GNSS noise is $\leq$ these figures.

- **Accuracy (closeness to true position) — not measurable in this dataset.** We have no surveyed reference point, and the tracks are of moving boats. The only accuracy proxy is each receiver's self-reported `hacc`, which is an estimate, not truth — and the two chips compute it from **different sentences** (LG290P from NMEA `GST`, LC29HEA from `PQTMPE`), so their `hacc` values are **not directly comparable**.

Results

Logging continuity (10 Hz target)

	Rate	Late/missed fix (dt>0.15 s)	Gaps > 1 s	Max gap
B2 (LC29HEA)	10.0 Hz	1247 (0.65 %)	0	0.3 s
E3 (LG290P)	10.0 Hz	239 (0.27 %)	0	0.4 s
E4 (LG290P)	10.0 Hz	1 (0.00 %)	0	0.3 s

All three hold a true 10 Hz with no dropouts. E4 was essentially flawless; B2 had the most late fixes but still 99.35 % on-cadence.

Signal quality

	Sats (med / min–max)	HDOP (med / p95 / max)	Fix quality
B2 (LC29HEA)	30 / 17–39	0.45 / 1.21 / 1.91	99.8 % SBAS, 0.2 % GPS
E3 (LG290P)	34 / 26–37	0.31 / 0.35 / 0.69	93.4 % SBAS, 6.6 % GPS
E4 (LG290P)	34 / 28–38	0.30 / 0.34 / 0.55	99.2 % SBAS, 0.8 % GPS

The **LG290P tracks ~4 more satellites and holds a markedly lower and more stable HDOP** (better sky geometry). The LC29HEA's HDOP is fine but wanders higher (up to ~1.9), consistent with its lower sat count.

Self-reported horizontal accuracy (`hacc`) — read with care

	median	mean	p95	max	varies?
B2 (LC29HEA)	2.17 m	2.32 m	3.63 m	10.6 m	yes — 3788 distinct, $\sigma=0.77$
E3 (LG290P)	1.30 m	1.31 m	1.41 m	2.07 m	barely — pinned near 1.30 m, $\sigma=0.05$
E4 (LG290P)	1.30 m	1.31 m	1.36 m	1.88 m	barely — pinned near 1.30 m, $\sigma=0.04$

⚠ **The LG290P's `hacc` is effectively a fixed ~1.30 m floor**, not a live, condition-tracking estimate — so “E3 1.3 m beats B2 2.2 m” is **not** a valid accuracy comparison. The LC29HEA's `hacc` genuinely varies with conditions and is the more informative (and more conservative) self-estimate.

Position precision — stillest-window scatter (the real measurement)

	Window	Mean SOG	σ -E / σ -N	DRMS	CEP50	CEP95	hacc in window
B2 (LC29HEA)	6428 s	0.01 kt	1.77 / 1.57 m	2.36 m	2.09 m	4.22 m	3.20 m
E3 (LG290P)	585 s	0.03 kt	0.92 / 0.90 m	1.29 m	0.88 m	3.00 m	1.31 m
E4 (LG290P)	318 s	0.03 kt	0.21 / 2.35 m	2.36 m	2.07 m	4.60 m	1.30 m

- **E3 (LG290P) is the cleanest:** ~0.9 m median radial scatter, symmetric (σ -E $\approx$ σ -N), over a long still window — that’s a genuine sub-metre precision result.
- **B2 (LC29HEA)** sat still the longest (107 min at 0.01 kt) and shows isotropic ~2 m scatter — a trustworthy ~2 m precision figure.
- **E4 (LG290P)** looks worse (2.4 m DRMS) **only because its scatter is wildly asymmetric** (σ -N 2.35 m vs σ -E 0.21 m) — that’s a boat swinging N-S on a mooring, not receiver noise. Same chip as E3, so its true precision is very likely ~E3’s; the number here is dominated by real motion. **Discount it.**
- In every case the measured scatter is **tighter than the receiver’s own hacc**, i.e. both chips report conservatively.

On sample size: there is exactly **one clean precision result per module type** (E3 for the LG290P, B2 for the LC29HEA), on **different boats with different antenna mounts and sky**. So these numbers characterise *this install of each module*, not the silicon in isolation — read “the LG290P unit on Wizard” rather than “the LG290P chip.” Satellite/HDOP counts (from both E3 and E4) are firmer since they agree across the two LG290P units.

Conclusions

1. **Stronger standalone GNSS front end on the LG290P installs (E-series).** Both LG290P units (E3, E4) tracked more satellites (~34 vs ~30) and held a lower, steadier HDOP (~0.30 vs ~0.45) than the LC29HEA (B2). Where precision was cleanly measurable — E3 — it reached sub-metre (~0.9 m CEP50); the LC29HEA (B2) install shows a solid ~2 m.
2. **Do not compare the two on hacc** — the LG290P pins it at a 1.30 m floor; the LC29HEA reports a live estimate. Different statistic, different sentence.
3. **Absolute accuracy was not measured** (no ground truth). To claim accuracy, do a known-point test (park each unit over a surveyed mark and compare) or enable **RTK**, which is the real accuracy lever — the RTK path targets ~cm and would dwarf these standalone differences.
4. **B1/Amigo is a reliability failure, not a data point** — it recorded a single fix and missed the race. Root-cause its logging (SD / power / config) before the next event.

Method / reproducibility

- **SailFrames units:** `scripts/gps_accuracy_race6.py` over `raw/<device>/2026-07-01/*_nav.csv` (ms, utc, lat, lon, alt, sog, cog, sat, hdop, fix, gps_date, hacc).
- **Owner-supplied tracks:** `scripts/gps_accuracy_race6_external.py` over the GPX files, the Vakaros `.vtx` (parsed with the handler’s `parse_vtx_full`), and the Sensor Logger `Location.csv`

exports (`horizontalAccuracy` → `hacc`).

- Common method: **stillest window** = longest contiguous run with SOG < 0.5 kt (≥ 30 s); scatter (σ -E/ σ -N, DRMS, CEP50/95) computed in a local E/N metric plane about the window mean. UTC unwrapped across the 00:00 rollover.

All nine tracks are linked into race `003bef01` and visible on the replay (SailFrames via `session_path`, owner tracks via `gpx_path`).